732A98 Visualization
Division of Statistics and Machine Learning
Department of Computer and Information Science
Linköping University

Computer lab 1

Instructions

- The lab report should be submitted via LISAM before the deadline. The submission should be an RMD file, HTML file and all supporting files compressed into one ZIP archive.
- Create a report for the lab solutions in RMarkdown. Make sure that it can be compiled to HTML and that all paths in RMD file are relative to the current directory where the RMD file is located. **Reports that can not be compiled are returned without revision.**
- **Include all your codes as an appendix into your report; Also show the corresponding parts of the code in the flowing text of the report.**
- A typical lab report should contain 2-4 pages of text plus some figures and the appendix with the code.
- Be concise and do not include unnecessary printouts and figures produced by the software that are not required in the assignments.

Assignment 1

The SENIC dataset describes the results of measurements taken at different US hospitals. An overview of the variables is given in the accompanying document **SENIC.pdf.**

1. Read the data from SENIC.txt into R.
2. Create a function that for a given column (vector) X does the following:
    a. It computes first and third quartiles Q1 and Q3 with quantiles()
    b. It returns indices of outlying observations, i.e. observation with X-values greater than Q3+1.5(Q3-Q1) or less than Q1-1.5(Q3-Q1).
3. Use **ggplot2** and the function from step 2 to create a density plot of *Infection risk* in which outliers are plotted as a diamond symbol (◊). Analyze the graph.
4. Produce graphs of the same kind as in step 3 but for all other quantitative variables in the data (to iterate over several column names, you can use .data, i.e. *aes(x=.data[variable_that_contains_column_name_as_string], …))*. Arrange these graphs into one (hint: use *arrangeGrob()* from the gridExtra package) and make some analysis.
5. Create a **ggplot2** scatter plot showing the dependence of *Infection risk* on the *Number of Nurses* where the points are colored by *Number of Beds*. Is there any interesting information in this plot that was not visible in the plots in step 4? Are there possible dangers of having such a color scale?
6. Convert the graph from step 3 to **Plotly** with the *ggplotly* function. What important new functionality have you obtained compared to the graph from step 3? Make some additional analysis of the new graph.

7. Create a **plotly** data plot-pipeline and the pipeline operator to make a histogram of *Infection risk* in which outliers are plotted as a diamond symbol (◊). **Hint**: *select(), filter()* and *is.element()* functions might be useful here.
8. Write a **Shiny** app that produces the same kind of plot as in step 4 but also includes:
    a. Checkboxes indicating for which variables the density plots should be produced
    b. A slider changing the bandwidth parameter in the density estimation (i.e., *'bw'* parameter)

    Comment on how the graphs change with varying bandwidth. Is there an optimal bandwidth for all graphs?

Assignment 2

In this assignment, you will analyze the mean incomes of Swedish households. Go to http://www.scb.se and choose "English" as language. Next, type "Disposable income for households by region, type of households and age" in the search bar and click "Search". Then click at "Statistical Database" and select "Disposable income for households by region, type of households and age. Year 2011 – 2023". In the database, select "Mean value, SEK thousands", all counties, age groups 18-29, 30-49 and 50-64, and year 2023. Download the "Comma delimited without heading" file.

1. Download a relevant map of Sweden with its counties from http://gadm.org/country and load it into R. Read your data into R and process it in such a way that different age groups are shown in different columns. Let's call these groups Young, Adult and Senior.
2. Create a **plotly** plot with three violin graphs showing the mean income distributions per age group. Analyze this plot and interpret your analysis in terms of income.
3. Create a surface plot in **plotly** showing the dependence of Senior incomes on Adult and Young incomes in various counties. What kind of trend can you see and how can this be interpreted? Do you think that linear regression would be suitable to model this dependence?
4. Use the *plot_geo()* function with trace "*choropleth*" to visualize the incomes of Young and Adults in two choropleth maps. Analyze these maps and make conclusions. Is there any new information that you could not discover in previous plots?
5. Use GPVisualizer http://www.gpsvisualizer.com/geocoder/ and extract the coordinates of Linköping. Add a red dot to the choropleth map for Young from step 4.

Assignment 3

In this assignment you will analyze tweets containing true and fake news about the COVID19 pandemic from the CIC truth seeker dataset 2023 (https://www.unb.ca/cic/datasets/truthseeker-2023.html). The *tweets.csv* file contains the text of the tweets and a label (i.e., 0 (fake) and 1 (true).

1. Create two word clouds, one for tweets containing fake news and one for real news. Make sure that stop words are removed. Which words are mentioned most often?
2. Aggregate data in chunks of 5 lines and compute sentiment values (by using “afinn” database) for fake and real news. Produce plots visualizing the aggregated sentiment values versus chunk index and make a comparative analysis between these plots. Does sentiment analysis show a connection of the corresponding documents to the kinds of text we expect to see in them?
3. Create the phrase nets for the real and fake news with the connector words “am”, “is”, “are”, “was”, and “were”. Analyze and compare both graphs.

Submission procedure

When submitting the report, remember to **specify in LISAM your group name** and all group members that wrote the report!

Make sure that you or your group mate submits the group report using *Lab X* item in the *Submissions* folder before the deadline. Make sure that **the report contains a Statement of Contribution** describing how each group member has contributed into the group report.

